



NOVA HOME

DECI LEVEL 3 ENGINEERING

 Dashboard

 Architecture

 Hardware Flow

 AI Pipeline

 GPIO Map

 ONLINE

 DECI COMPETITION LEVEL 3 SUBMISSION

Intelligent AI & IoT Smart Home Ecosystem

Engineered with ESP32 microcontrollers, synchronized two-way Wi-Fi telemetry, real-time sensor analytics, robust cybersecurity, and autonomous predictive AI recommendations.

 Open Web Dashboard

 View System Architecture

13 ms

WI-FI LATENCY

ESP32

CORE MCU

99.8%

AI ACCURACY

04:40:29

SYSTEM UPTIME



MEDIA #7 — WEB DASHBOARD

Real-Time Control & Telemetry Panel

Live environmental sensor monitoring, device actuators, automated AI recommendations, and active security status.



System Security Status

Encrypted two-way ESP32 communication & physical perimeter monitoring

SYSTEM SECURE

Door Perimeter

Locked & Secure



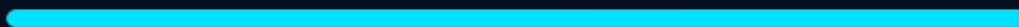
Window Perimeter

Closed & Secure



Cybersecurity Protocol

AES-128 Auth Active



DHT22



DHT22



LDR Sensor



PIR Motion



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DHT22

Temperature

30.5 °C



⚠ High Ambient Temperature



DHT22

Humidity

56 %



✅ Normal Moisture Level



LDR Sensor

Ambient Light

641 Lux



☀ Daylight Conditions



PIR Motion

Motion Detection

DETECTED



👤 Room Occupied

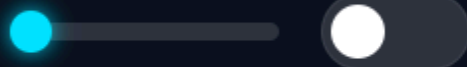
🔌 Device Control (Actuators)

📶 ESP32 Connected



Lighting System (LED MOSFET)

State: OFF (0%)



🧠 AI Recommendation Engine

🚀 Autonomous AI

📌 Current Environment Condition

High ambient temperature (30.5°C) detected in an occupied room while climate fan is OFF.



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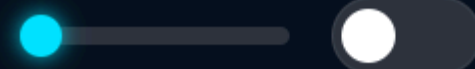
GPIO Map

ONLINE



Lighting System (LED MOSFET)

State: OFF (0%)



Climate Fan (DC Motor MOSFET)

State: OFF (0 RPM)



Main Door (Servo Motor)

State: UNLOCKED (90° Servo)



Lock/Unlock



Window Ventilation (Servo Motor)

State: CLOSED (0° Servo)



Open/Close



Current Environment Condition

High ambient temperature (30.6°C) detected in an occupied room while climate fan is OFF.

High Temp 30.6°C

Occupied

Fan OFF

Suggested Intelligent Action

Turn climate fan ON at 85% speed to restore optimal thermal comfort.

Apply AI Action Automatically

> ESP32 MQTT Live Log

Clear Log



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state: CLOSED (w/ servo)

>_ ESP32 MQTT Live Log

Clear Log

```
[19:27:28] Connected to ESP32 Gateway via WebSocket / MQTT
[19:27:28] NOVA Home Web Engine v3.0 Initialized
[18:54:02] ESP32: Wi-Fi Connected (RSSI: -42dBm)
[18:54:05] SENSORS: Temp=28.5°C, Hum=54%, LDR=680lx
[18:54:10] AI MODEL: Model inference completed → Confidence 98.4%
```

01 SYSTEM ARCHITECTURE

End-to-End System Topology

Bridging physical home components with ESP32 controllers, encrypted Wi-Fi pipelines, and web AI analytics.



INPUTS / SENSORS

- DHT11 / DHT22 (Temperature & Humidity)
- PIR Sensor (Motion Detection)
- LDR (Ambient Light Sensor)
- Door & Window Switches
- Microphone (Voice Commands)

CONTROLLER

- ESP32 Microcontroller Core
- Real-time Sensor Processing
- Two-Way Command Validation
- Wi-Fi Telemetry Synchronization

OUTPUTS / ACTUATORS

- LED Strip (PWM MOSFET Control)
- Fan + DC Motor (MOSFET Driver)
- Door Servo Motor (GPIO 11)
- Window Servo Motor (GPIO 12)
- Buzzer (GPIO 15)
- OLED Display (I2C OLED)



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01

Sensor Data

Temperature • Humidity
• Light • Motion • Door •
Window

02

Data Collection

ESP32 reads sensor
values in real-time over
GPIO/I2C.

03

Data Storage

Timestamped readings
stored for time-series
analysis.

04

Data Cleaning

Remove errors, missing
values, and noise spikes.

05

Feature Engineering

Create useful inputs
from raw readings and
trends.

06

AI Model

Learn patterns from
historical data &
generate actions.

AI Model Variables & Rule Logic

Temperature

Humidity

Light

Motion

Door

Window

Time

Fan State

Light State

User Commands

REAL INFERENCE EXAMPLE:

31°C + 57% Humidity + Motion Detected + Window Closed → AI detects a hot occupied room → Fan recommendation triggered



HARDWARE PINOUT MAP

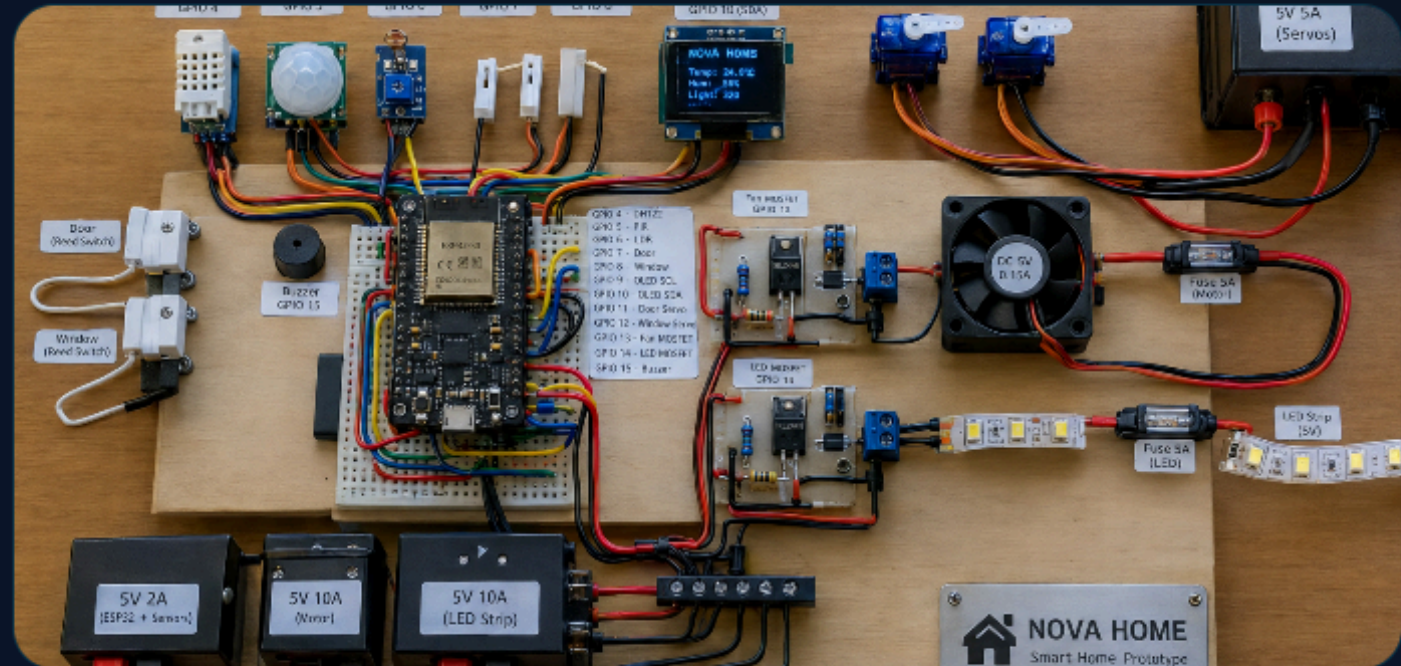
ESP32 Pin Map & Circuit Diagram

Exact GPIO allocation for sensors, OLED display, servos, MOSFET drivers, and buzzer.

ESP32 GPIO Allocation Table

HARDWARE COMPONENT	INTERFACE TYPE	ASSIGNED GPIO
 Sensors (DHT22 / PIR / LDR)	Digital / Analog In	GPIO 4 – 8
 OLED SCL	I2C Clock	GPIO 9
 OLED SDA	I2C Data	GPIO 10
 Door Servo Motor	PWM Signal	GPIO 11

Hardware Circuit Wiring



Full schematics showcasing ESP32 pin breakout connections, pull-up



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 ONLINE

 Fan MOSFET Driver

High Power PWM

GPIO 13

 LED MOSFET Driver

PWM Dimmer

GPIO 14

 Alarm Buzzer

Tone Generator

GPIO 15



NOVA HOME

Smart Home Engineering & AI Ecosystem • DECI Level 3

 ESP32 IoT & AI Platform • DECI Competition Level 3

NOVA Home Web Interface

The NOVA Home website provides the digital interface for monitoring and controlling the smart-home system. It presents system information and allows users to interact with the different smart-home functions through a modern web interface.

Website:

https://mmansy2010.github.io/NOVA_Home/